Dashboard Architecture & Planning

Module 09 — Day 1: Dashboard Planning & Application Setup

Customer Segmentation Project

1. Review of Previous Project Outputs

Item	Source	Identified Asset
Final dataset	Module 08	data/processed/customer_segments_with_strategy.csv (2,237 rows x 67 cols)
Final engineered features	Module 05	13 features: Customer_Age, Customer_Tenure, Total_Spending, Total_Purchases, etc.
Final segmentation model	Module 06	K-Means, K=4, StandardScaler-normalized, 12-feature vector
Saved model files	Module 06	models/production_pipeline.pkl, scaler.pkl, segmentation_model.pkl
Customer segment labels	Module 07	High-Value / Premium-Loyal / Discount Seekers / New-Developing
Business recommendations	Module 08	Marketing, product, pricing, retention, reactivation strategy per segment

2. Information Selected for the Dashboard

All information required by the module brief is included, organized into 10 sections (Section 2 below). Nothing was added purely for visual filler — every chart is tied to a specific business question, per the project's 'no charts just to increase chart count' rule.

3. Dashboard Navigation Structure

#	Section	Key Content
1	Executive Overview	Total customers, segments, avg/total spend, avg income, purchase frequency, campaign response
2	Customer Demographics	Age, income, education, marital status, children, household info
3	Spending Analysis	Total/avg spending, category spend, segment spend, purchase frequency
4	Product Analysis	Wine/Meat/Fruit/Fish/Sweets/Gold spend and segment preference
5	Purchase Channel Analysis	Web/Store/Catalog purchases, online activity
6	Campaign Response	6-campaign acceptance, response rate, complaints, by-segment performance
7	Customer Segments	Cluster sizes/%, characteristics, spending, income, activity
8	Segment Comparison	A vs B on income/spending/purchases/recency/response/value
9	Individual Customer Profile	ID lookup + live new-customer prediction form
10	Marketing Recommendations	Module 08 recommendations rendered per segment

4. Technology Selection

4.1 Options Considered

Option	Verdict
Streamlit (selected)	Connects directly to our Python/scikit-learn model; fastest to build and iterate on; well-suited to a small ML intern team

Option	Verdict
Power BI / Tableau	Strong for BI reporting, but does not natively run our live Python K-Means prediction pipeline
Plotly Dash	Capable alternative, but more boilerplate than Streamlit for the same result
Flask/FastAPI + custom frontend	Most flexible, but far more development time than justified for this project's scope

4.2 Decision

Streamlit is selected as recommended by the module brief: it loads the trained scikit-learn pipeline directly, requires no separate frontend build step, and lets the same Python feature-engineering module be reused for both training and live prediction.

5. Project Application Structure

```
customer-segmentation-project/ │ ├── app/ │   ├── app.py                # Entry point │
├── dashboard.py          # Assembles all 10 sections │   └── components/          #
filters.py, charts.py, customer_lookup.py, data_loader.py │ ├── data/ │   ├── raw/
# marketing_campaign.csv │   └── processed/            #
customer_segments_with_strategy.csv │ ├── models/ │   └── production_pipeline.pkl  #
scaler + K-Means + feature list │ ├── src/ │   ├── preprocessing.py      # Cleaning +
input validation │   ├── feature_engineering.py # Feature creation (shared
train/predict) │   └── prediction.py          # Loads model, assigns segment +
recommendation │ ├── tests/                    # pytest suite ├── reports/
# QA reports ├── notebooks/                 # Key notebooks ├── requirements.txt └──
README.md
```

6. Day 1 Deliverables Summary

- Dashboard architecture defined (this document).
- Project folder structure created and verified.
- Technology stack selected and justified: Streamlit.
- Final dataset, features, model, labels, and recommendations identified and located.
- requirements.txt and README.md drafted for the new application.